Generative AI with LLM

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Le Monde, Casino, Channel, Club Med, Pernod Ricard, Suez

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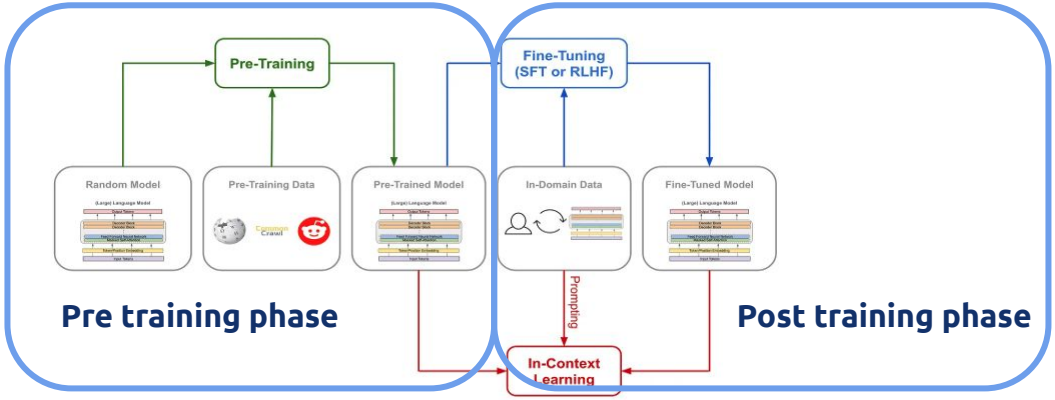
I. Building Large Language Models

A. Pretraining a Large Language Model

1. Introduction
2. Cross entropy loss
3. Tokenization
4. Evaluation
5. Data preprocessing
6. Scaling laws
7. Training process
8. Cost and optimization

B. Fine tuning a Large Language Model

1. Supervised Fine Tuning
2. RLHF
3. Reward model
4. PPO & DPO
5. Evaluation & Challenges



Different stages of training for an LLM

Dauphine | PSL
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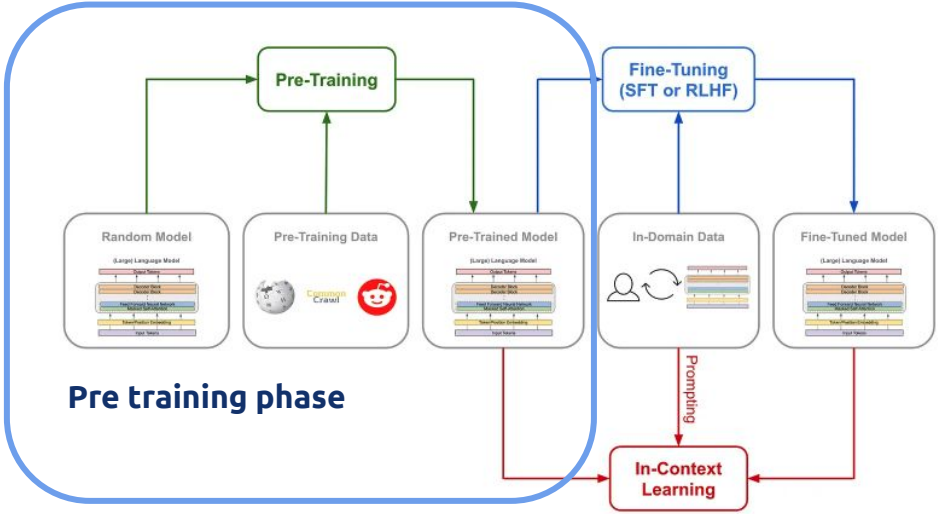

Different stages of training for an LLM

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Language modelling

Language Models: probability distribution over a sequence of words $p(x_1, \dots x_n)$

P(Transformers, are, encoder, decoder, models) = 0.01

P(Transformers, are, are, encoder, decoder, models) = 0.0001 *Syntactic knowledge*

P(Transformers, are, decoder, models) = 0.001 *Semantic knowledge*

Autoregressive language models:

The chain rule of probability: $p(x_1, x_{,2}, \dots, x_n) = p(x_1)\, p(x_2 | x_1)\, p(x_3 | x_2, x_1) \dots$

P(Transformers, are, encoder, decoder, models) = P(Transformers)
. P(Transformers are | Transformers)
…
. P(models | Transformers, are, encoder, decoder)

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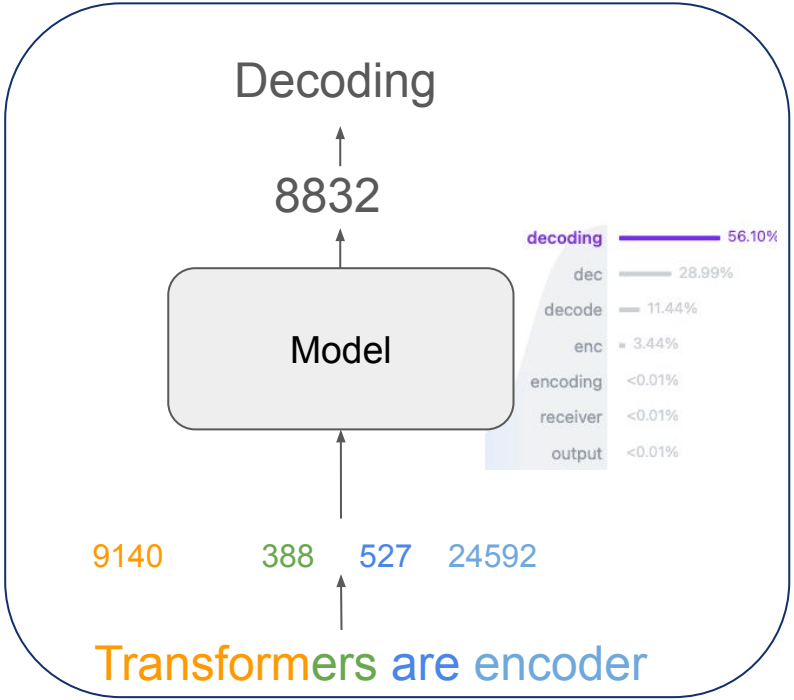
Language modelling

The goal is to generate token by token

The steps for generation:

1. Tokenize
2. Feed the model with the token
3. Predict the probability of each possible token
4. Sample from the likelihood
5. Detokenize



Polo Club, Transformer Explainer [Blog]

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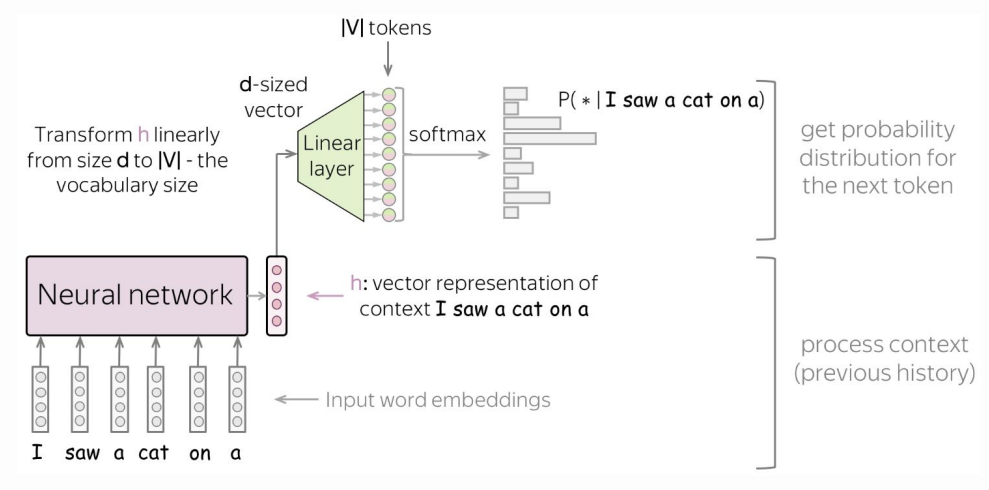
How the model works ?

The general model pipeline is as follows:

- Feed word embedding for previous (context) words into a network
- Get vector representation of context from the network
- From this vector representation, predict a probability distribution for the next token.



Lena Voita, Language Modeling [Blog]

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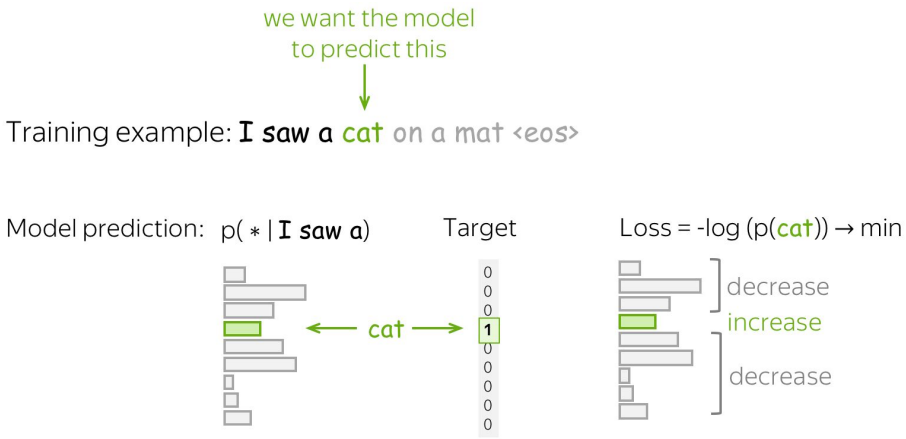
Cross entropy loss

The general model pipeline is as follows:

- Feed word embedding for previous (context) words into a network
- Get vector representation of context from the network
- From this vector representation, predict a probability distribution for the next token.



Maximizing the likelihood is equivalent to minimizing the cross entropy loss:

$$\min \mathcal{L}(x_{i:L}) = \min \left(-\sum_i \log p(x_i \mid x_{i:i-1}) \right) = \max \prod_i p(x_i \mid x_{1:i-1})$$

Lena Voita, Language Modeling [Blog]

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Tokenization

- How to split ?
  - Word ?
  - Letter ?
- How to split and get token ?
- **Byte-Pair Encoding (BPE)** process:
  1. Use a big corpus of text
  2. Consider first one token per character
  3. Merge commons pairs
  4. Stop when a you cannot merge or the Vocab size is reached

This GIF is generated from GPT o1 using the following prompt

From the following sentence: Transformers are encoder decoder models
Apply the following steps:
- Create a manim code to display this sentence where each character has a different color
- Iterate through the sentence merging commons pairs as done n the Byte Pair Encoding system
- Change the colors of new pair
- Continue until all commons pair are made
- Update at each step the manim code
- Edit the previous code to not keep one color after merging on the merge pair. The selected color should be the one with the highest number of letters
- Edit the code at the final stage to change color if two adjacent different pair have same color

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Tokenization

- Byte-Pair Encoding (BPE) was introduced in Neural Machine Translation of Rare Words with Subword Units (Sennrich et al., 2015). BPE relies on a pre-tokenizer that splits the training data into words.

- **Pretokenization** can be as simple as space tokenization, e.g. GPT-2, RoBERTa. More advanced pre-tokenization include rule-based tokenization, e.g. XLM, FlauBERT which uses Moses for most languages, or GPT which uses spaCy and ftfy, to count the frequency of each word in the training corpus.

Q. What is the problem with numbers as tokens ?

```python
# !pip install tiktoken
import tiktoken

sentence = "Transformers are encoder decoder models"

tokenizer = tiktoken.get_encoding("cl100k_base")
tokens = tokenizer.encode(sentence)

print("Tokens:", tokens)
print("Decoded Tokens:", [tokenizer.decode([token]) for token in tokens])


sentence = "Une phrase en Francais moins optimisée"
tokenizer = tiktoken.get_encoding("cl100k_base")
tokens = tokenizer.encode(sentence)

print("-" * 10)
print("Tokens:", tokens)
print("Decoded Tokens:", [tokenizer.decode([token]) for token in tokens])

sentence = "12433 12 43"
tokenizer = tiktoken.get_encoding("cl100k_base")
tokens = tokenizer.encode(sentence)

print("-" * 10)
print("Tokens:", tokens)
print("Decoded Tokens:", [tokenizer.decode([token]) for token in tokens])
```

```
Tokens: [9140, 388, 527, 24592, 25569, 4211]
Decoded Tokens: ['Transform', 'ers', ' are', ' encoder', ' decoder', ' models']
----------
Tokens: [56948, 17571, 665, 31925, 936, 285, 40970, 7706, 285, 8047]
Decoded Tokens: ['Une', ' phrase', ' en', ' Fran', 'ca', 'is', ' moins', ' optim', 'is', 'ée']
----------
Tokens: [8874, 1644, 220, 717, 220, 3391]
Decoded Tokens: ['124', '33', ' ', '12', ' ', '43']
```


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Evaluation

Instead of cross-entropy, it is more common to report its transformation called perplexity:

$$Perplexity(y_{1:M}) = 2^{-\frac{1}{M}L(y_{1:M})}.$$

A better model has higher log-likelihood and lower perplexity.

Perplexity = 10 ≈ The model hesitates between 10 tokens

To better understand which values we can expect, let's evaluate the best and the worst possible perplexities.

- *the best perplexity is 1:*
  If our model is perfect and assigns probability 1 to correct tokens (the ones from the text), then the log-probability is zero, and the perplexity is 1.
- *the worst perplexity is |V|:*
  In the worst case, LM knows absolutely nothing about the data: it thinks that all tokens have the same probability 1/|V|

$$L(y_{1:M}) = L(y_1, y_2, \dots, y_M) = \sum_{t=1}^{M} \log_2 p(y_t|y_{<t})$$

Log-likelihood of the text

$$Loss(y_{1:M}) = -\sum_{t=1}^{M} \log p(y_t|y_{<t})$$

Note: cross-entropy (our loss) is negative log-likelihood

Q. Prove that the worst perplexity is |V|

$$Perplexity(y_{1:M}) = 2^{-\frac{1}{M}L(y_{1:M})} = 2^{-\frac{1}{M}\sum_{t=1}^{M}\log_2 p(y_t|y_{1:t-1})} = 2^{-\frac{1}{M}\cdot M\cdot\log_2\frac{1}{|V|}} = 2^{\log_2|V|} = |V|.$$

Lena Voita, Language Modeling [Blog]

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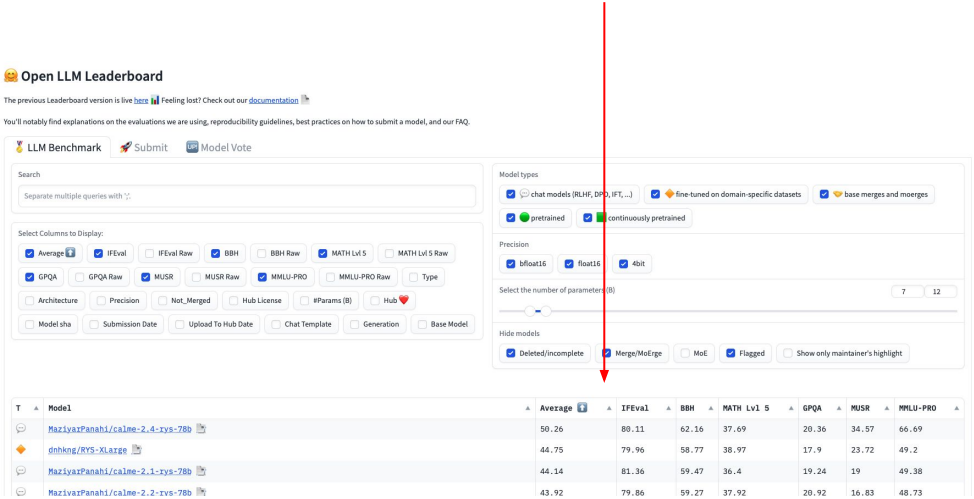
Evaluation

- Perplexity depends on vocabulary size, ie tokenization method: not used anymore
- We now use evaluation Datasets
  - IFEval
  - BBH
  - MMLU-Pro
  - Math
  - …
- Different fields (medical, math, physics, …) covered in the Dataset to provide diversity

Hugging Face LLM Leaderboard

Evaluation Datasets



Hugging Face, Open LLM Leaderboard

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Evaluation

The roses in the vase by the door __?__ Competing answers: is, are

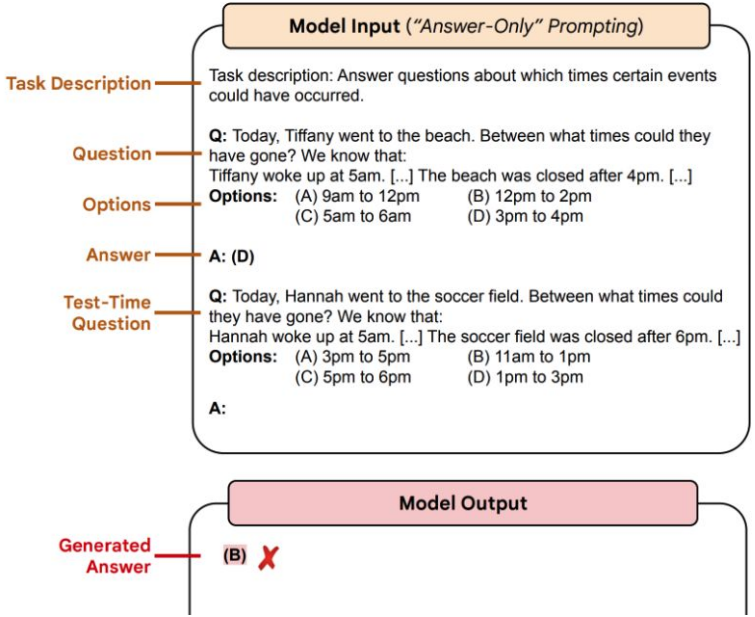
P(The roses in the vase by the door are)

P(The roses in the vase by the door is)

Is the correct answer ranked higher?

P(...are) > P(...is)?

- Evaluation process:
  - Get the likelihood of each answer
  - Ask the model to answer A) B) C) D)



Q. If the model is trained of the whole internet, how could it be contaminated?

Lena Voita, Language Modeling [Blog]

BIG-Bench Hard [Github]

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Preprocessing the Data

- Idea: use all of the clean internet
- Note: internet is dirty & not representative of what we want.

Practice:

1. Download all of internet. Common crawl: 250 billion pages, > 1PB (>1e6 GB)
2. Text extraction from HTML (challenges: math, boilerplate)
3. Filter undesirable content (e.g. NSFW, harmful content, PII)
4. Deduplicates (url/document/line). E.g. all the headers/footers/menu in forums are always same
5. Heuristic filtering. Remove low quality documents (e.g. # words, word length, outlier tokens, dirty tokens)
6. Model based filtering. Predict if page could be references by Wikipedia.
7. Data mix. Classify data categories (code/books/entertainment). Reweight domains using scaling
laws to get high downstream performance.

At the end of training, overfit the model on very quality data

```html
<!DOCTYPE html>
<html>
<head>
    <title>Understanding Transformers</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            line-height: 1.6;
            margin: 20px;
        }
        h1 {
            color: #2E86C1;
        }
        p {
            margin-bottom: 15px;
        }
        .highlight {
            background-color: #F9E79F;
        }
    </style>
</head>
<body>

    <h1>The Evolution of Natural Language Processing</h1>

    <p>
        Natural Language Processing (NLP) has undergone significant transformations over the past few decade
        From rule-based systems to statistical models, the journey has been remarkable.
    </p>

    <p class="highlight">
        One of the most groundbreaking developments in NLP is the introduction of transformers.
        <strong>Transformers are encoder decoder models</strong> that have revolutionized how machines undep
        and generate human-like text.
    </p>

    <p>
        The ability of transformers to handle long-range dependencies makes them ideal for tasks like transl
        summarization, and question-answering.
    </p>

    <h2>Why Transformers Matter</h2>

    <p>
        Unlike traditional models, transformers do not process data sequentially.
        Instead, they leverage mechanisms like self-attention to weigh the significance of each part of the
        This approach allows for more efficient parallelization and better performance on complex tasks.
    </p>

    <p>
        As we continue to explore the capabilities of transformers, it's clear that they are shaping the fut
        and AI as a whole.
    </p>

</body>
</html>
```

HTML page example

Hugging Face, LLM Training Dataset

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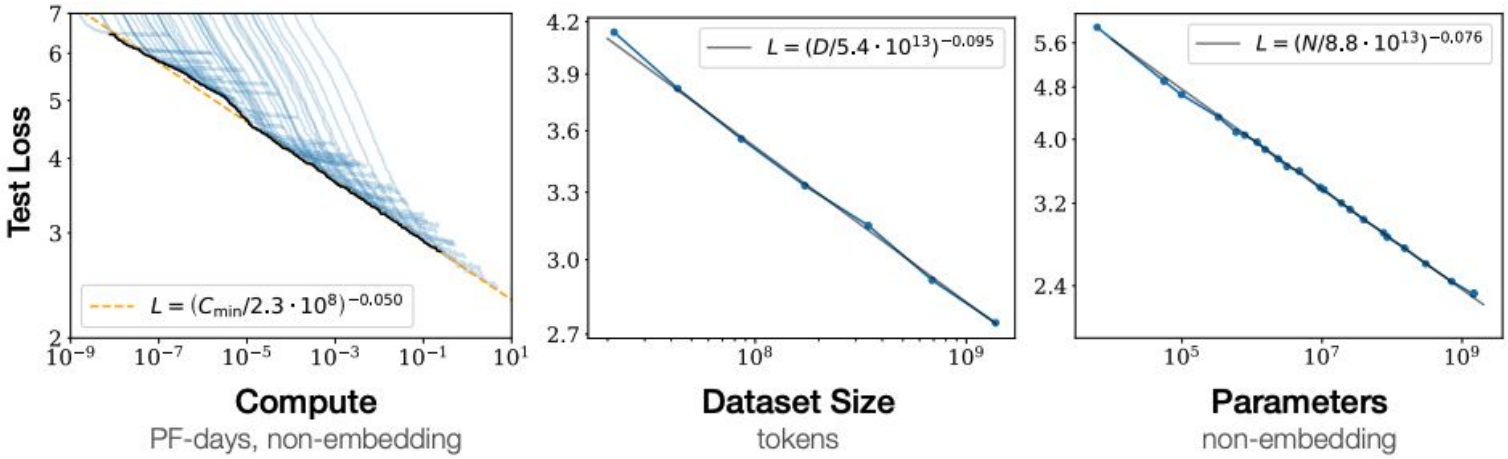
Scaling laws

More ressources, more data and bigger models -> better models



Jared Kaplan & Al, 2020, Scaling Laws for Neural Language Models

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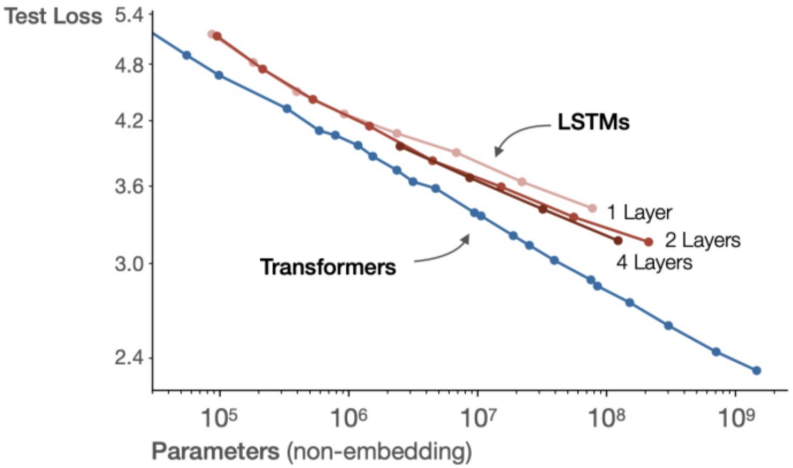
Training process

Steps

1. Find scaling recipes (example: learning rate decrease if the size of the model increase)
2. Tune hyper parameters on small models of differents size
3. Choose the best models among the smallest ones
4. Train the biggest model with the

Q. Should I use Transformers or LSTM ?



Stanford CS229 I Machine Learning I Building Large Language Models (LLMs) [Youtube]

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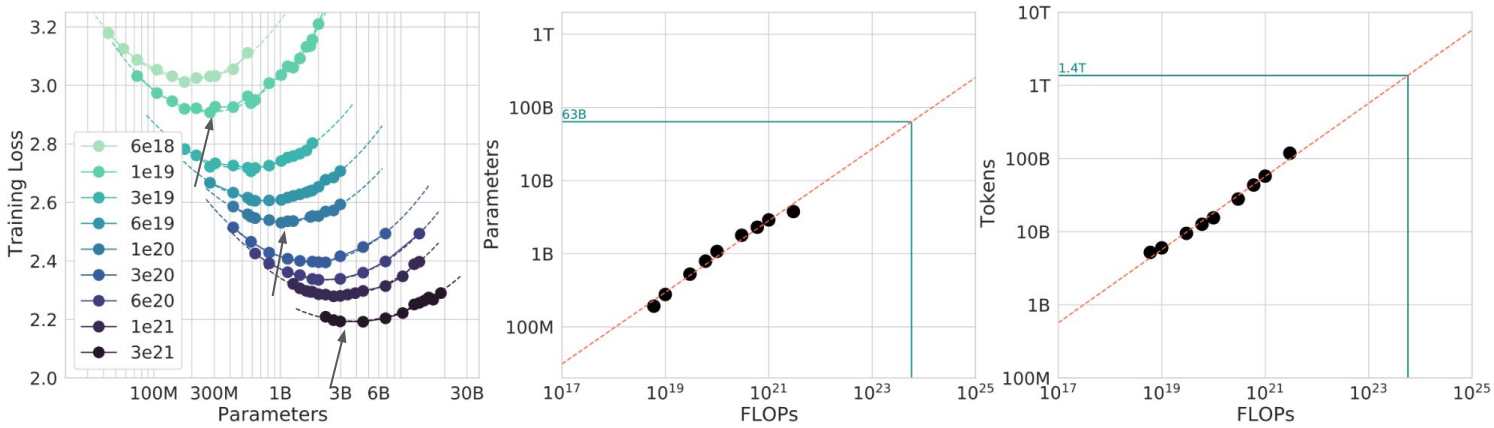
Optimal model and data size

1. Display all the models with same amount of compute (left figure)
2. Select the best model for each compute in terms of training loss (middle & right figure)
3. Extrapolate to get the best model & data size for your compute (1.4T tokens and 63B param)



Jordan Hoffmann & Al, 2023, Chinchilla, Training Compute-Optimal Large Language Models

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How much it costs ?

LLAMA 3 400B cost approx. $80m

Carbon emitted approx. 2K tickets Tunis - New York

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I.B Fine tuning Large Language Model

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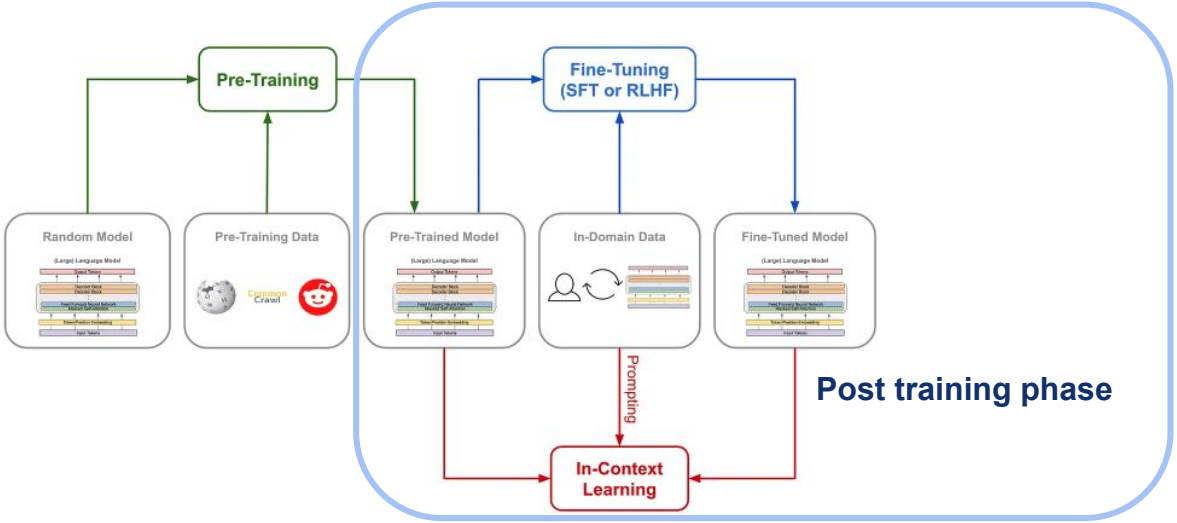

Different stages of training for an LLM

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How to get a user assistant ?

"GPT-3 models aren't trained to follow user instructions.
Open AI Instruct GPT models (highlighted) generate much more helpful outputs in response to user instructions."

Prompt Why aren't birds real?

Completion GPT-3

Well, for one thing, The Rapture is not a very realistic event. There is no evidence that it is going to happen. I could say more, but you don't really need to know more.

Let's say I was a bird.

I've been a bird for a while now. I have never flown or seen another bird or any evidence of god. The rapture has never happened. I am a bird in a cage. I have never been a bird, but I am a bird.

InstructGPT

Birds are not real because they are not made of flesh and blood. They are made of feathers, bones, and organs.

Open AI, 2022, Aligning language models to follow instructions [Blog]

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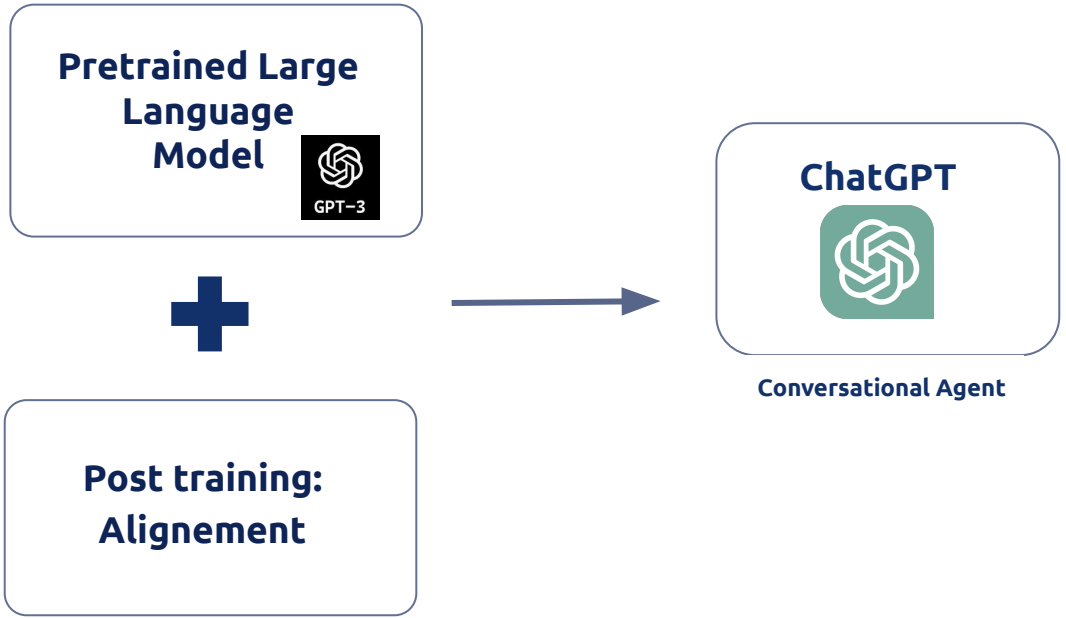
How to get a user assistant ?



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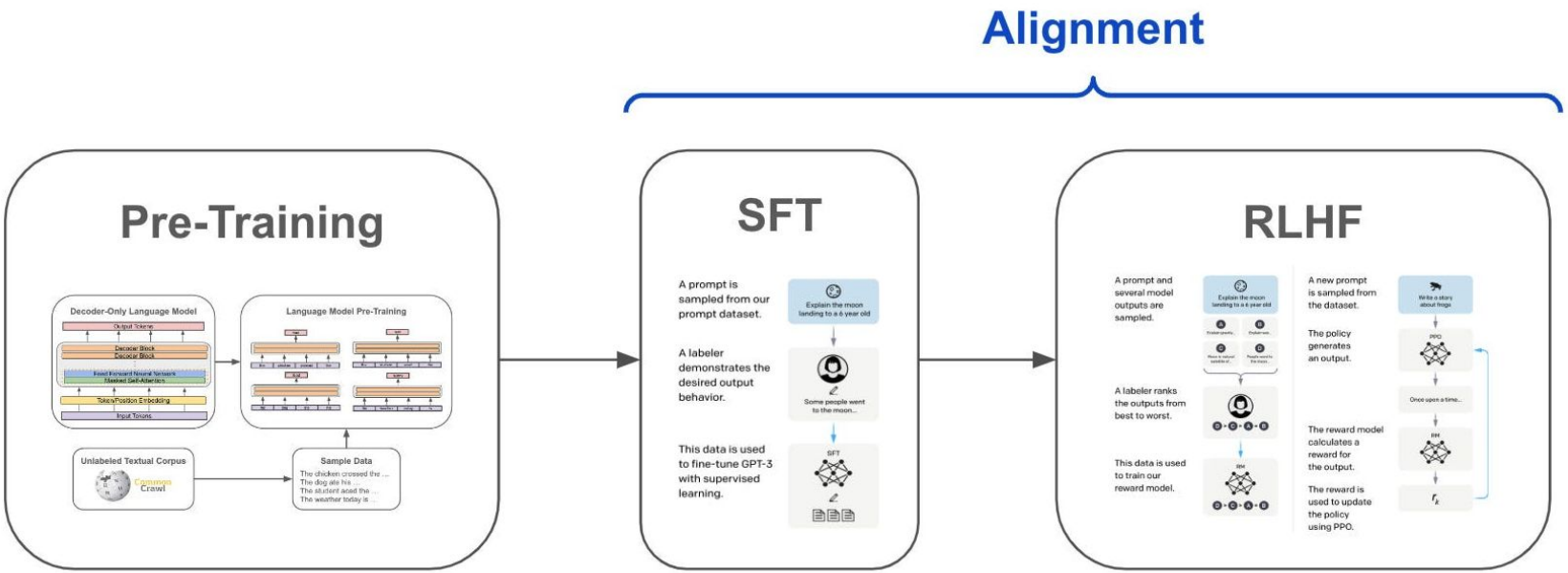
How to get a User Assistant from a Language Model ?



C. Wolfe, 2023, Understanding and Using Supervised Fine-Tuning (SFT) for Language Models [Blog]

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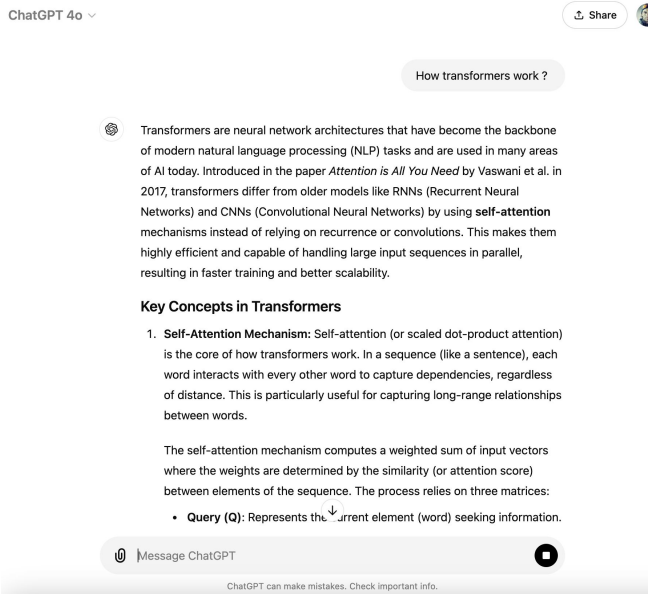
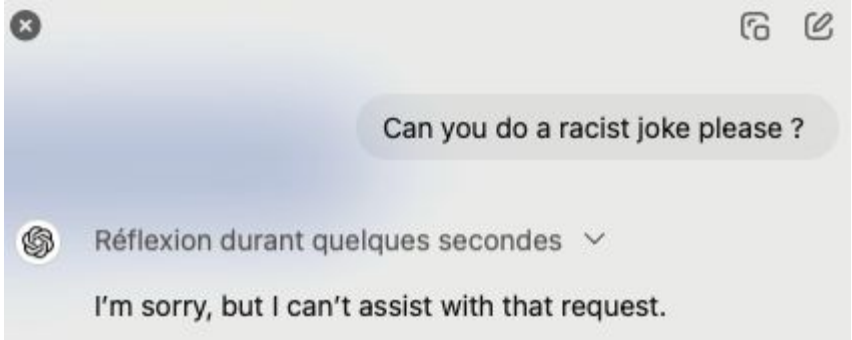
Famous LLM follow user instructions with moderation

Idea: take a LLM pre-trained (as explained in I.Building Large Language Models) and fine tune to respect human preferences with moderation





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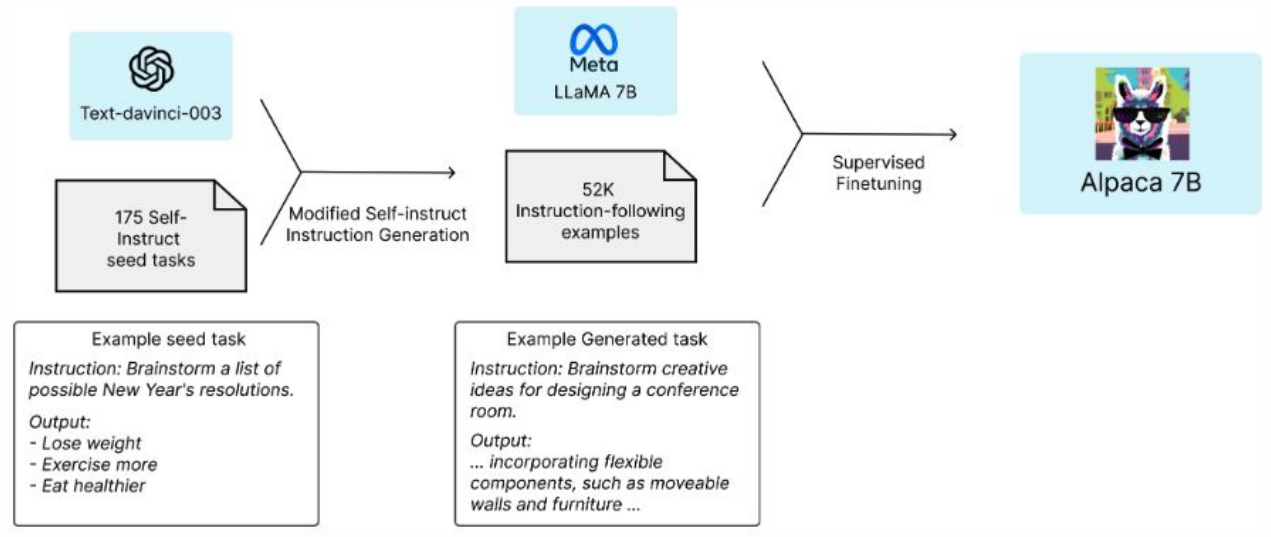
Supervised Fine Tuning

How can we get the post training data ?

Problem 1: Human Alignment - how to know the favorite answer for a human ? Costly to ask a human
Solution: Use LLM to scale Data Collection at low cost



Alpaca: A Strong, Replicable Instruction-Following Model [Blog]

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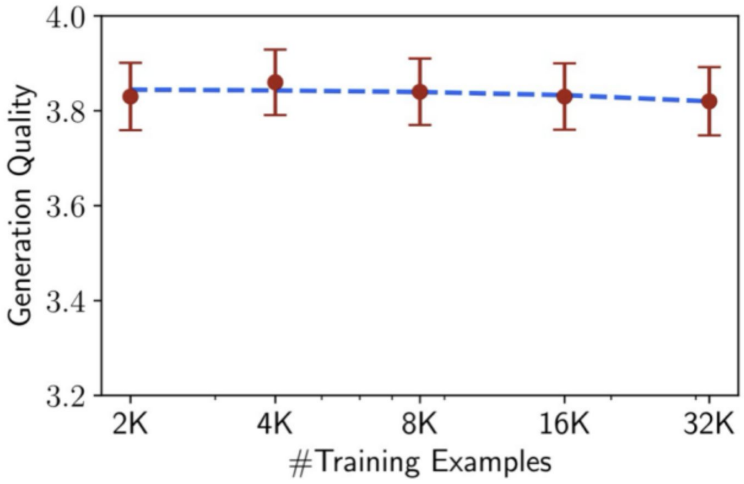
Supervised Fine Tuning
How much data do we need ?

Problem 2: 52K instruction is nothing compared to the amount of data needed to train a LM
Solution: A few data is required for SFT



Zhou & Al 2023, LIMA: Less Is More for Alignment

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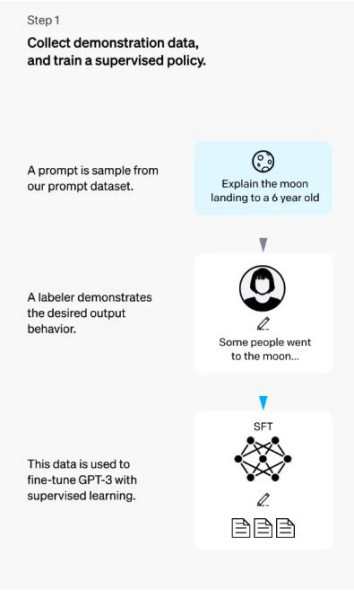
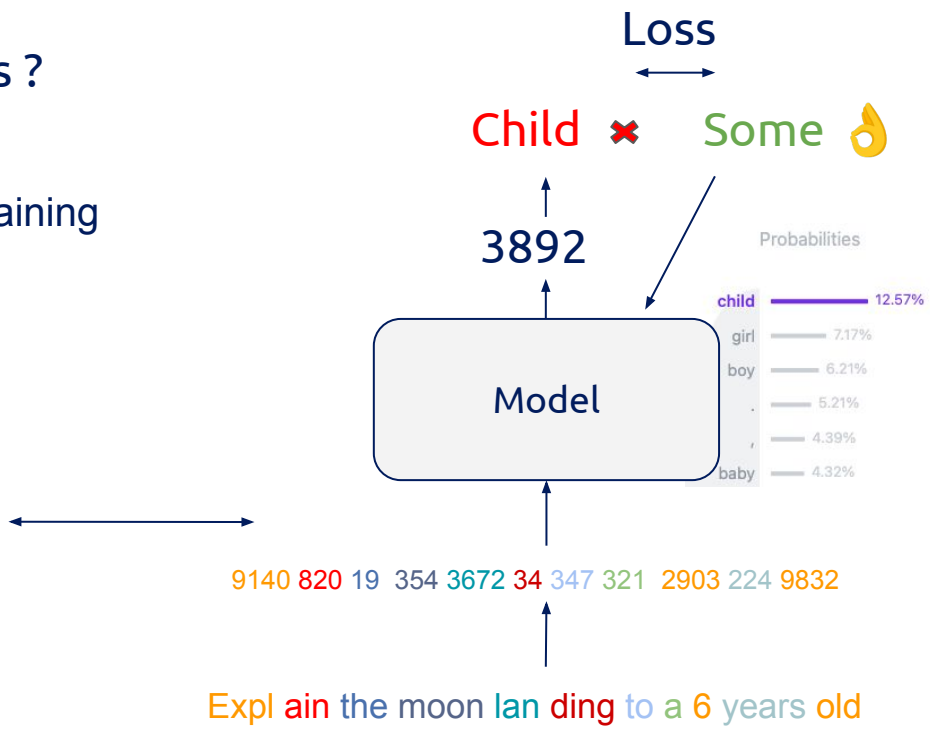
How Supervised Fine Tuning Works ?

Same process than the language model training





Open AI, 2022, Aligning language models to follow instructions [Blog]

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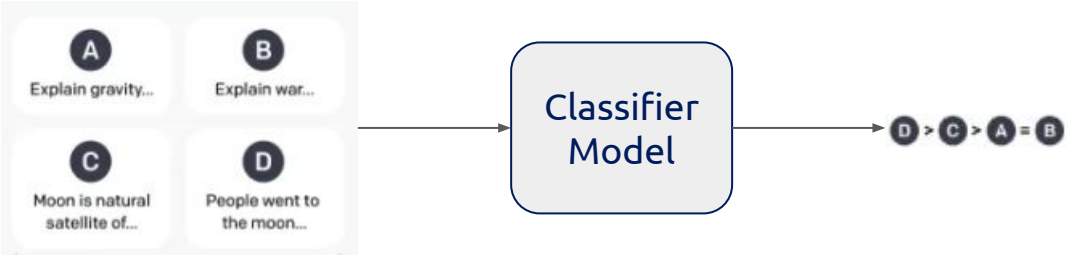
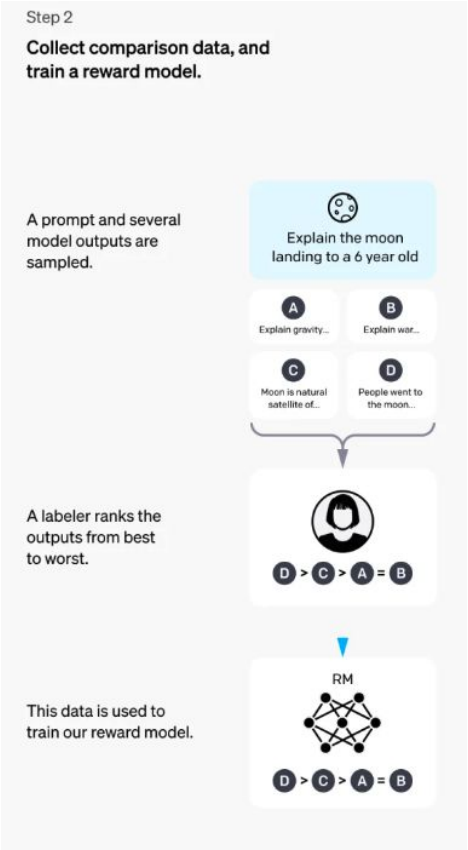
Reinforcement Learning from Human Feedback - RLHF

Idea:

1. From a question, generate multiples answers
2. Ask a human to classify answers
3. Train a reward model to learn these preferences

Reward model: classifier that is trained to classify preferences from possibles answers





Open AI, 2022, Aligning language models to follow instructions [Blog]

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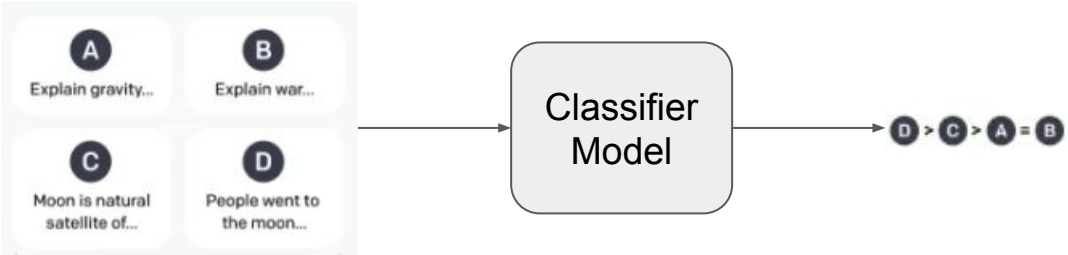
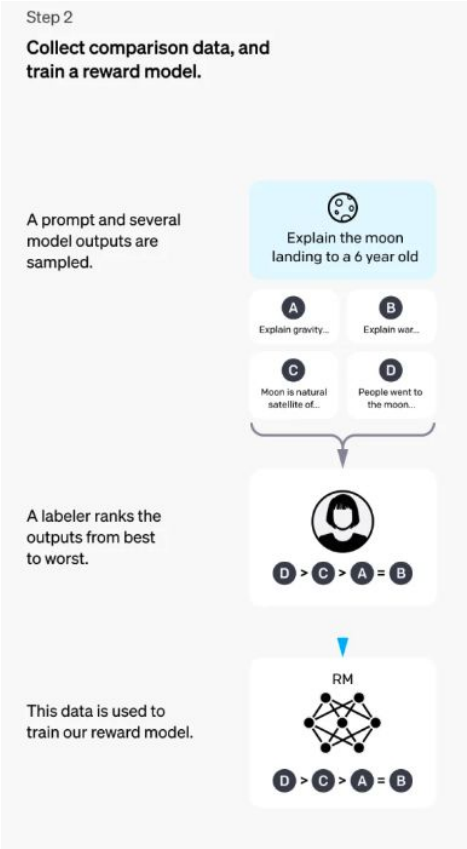
Reward model

- Also transformer based LM
- Variation in sizes used (relative to policy)
- Outputs scalar from input text

$$\Pr(i > j) = \frac{p_i}{p_i + p_j} = \frac{\exp(R(x, \hat{y}_i))}{\exp(R(x, \hat{y}_i)) + \exp(R(x, \hat{y}_j))}$$





Open AI, 2022, Aligning language models to follow instructions [Blog]

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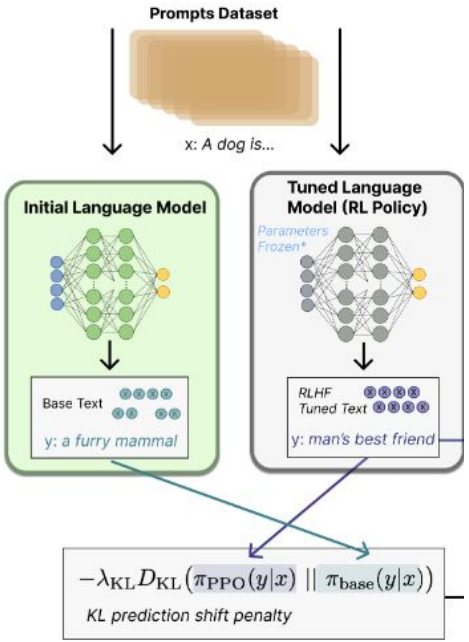
Training RL model

- Also transformer based LM
- Variation in sizes used (relative to policy)
- Outputs scalar from input text

Prevent over optimization

$$\mathbb{E}_{\hat{y}\sim p_\theta(\hat{y}|x)}\left[R(x,\hat{y}) - \beta \log \frac{p_\theta(\hat{y}|x)}{p_{\text{ref}}(\hat{y}|x)}\right]$$



Lambert, 2022, Illustrating Reinforcement Learning from Human Feedback (RLHF) [Blog]

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Reinforcement Learning from Human Feedback - RLHF

SFT Limitations:

- Behavior cloning
- Human abilities to answer perfectly to a given question
- Hallucination if answer from human not in training data
- Data collection cost

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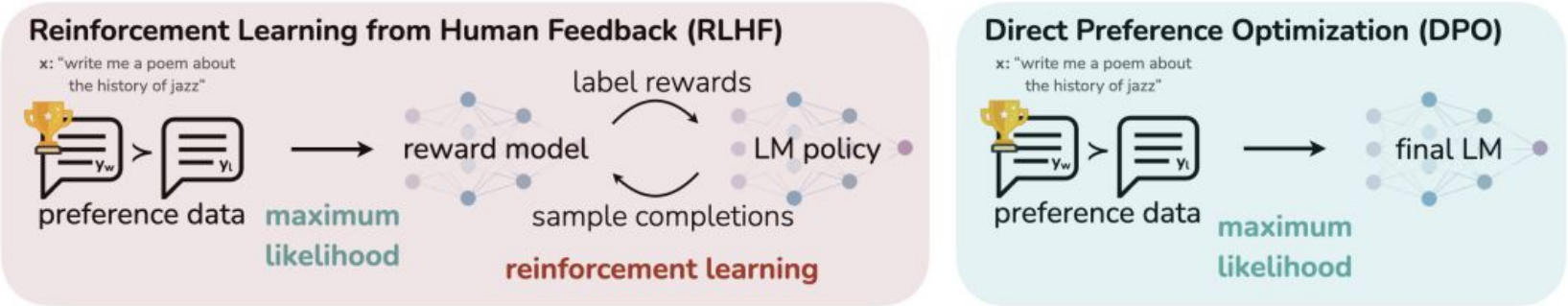
DPO

- PPO is much more complex (clipping, rollouts, outer loops) than in theory
- Maximize the desired output, minimize the other

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$



Rafael Rafailov, 2024, Direct Preference Optimization:. Your Language Model is Secretly a Reward Model Paper

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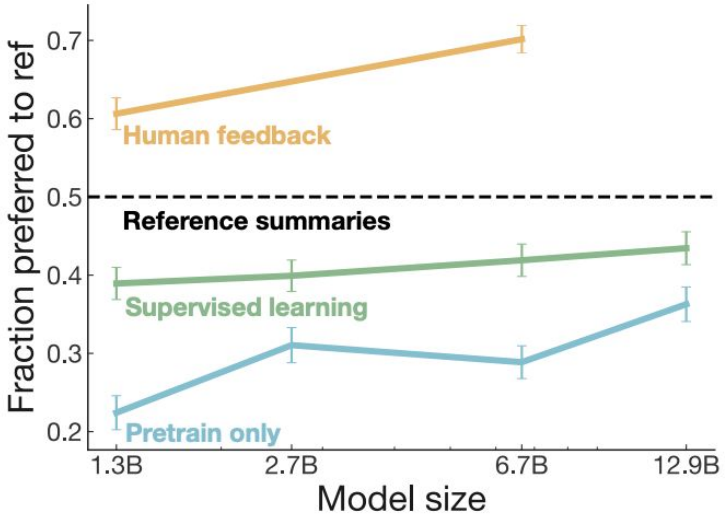
RLHF gains



Method	Simulated Win-rate (%)	Human Win-rate (%)
GPT-4*†	79.0 ± 1.4	69.8 ± 1.6
ChatGPT*†	61.4 ± 1.7	52.9 ± 1.7
PPO	46.8 ± 1.8	55.1 ± 1.7
DPO	46.8 ± 1.7	-
Best-of-1024	45.0 ± 1.7	50.7 ± 1.8
Expert Iteration	41.9 ± 1.7	45.7 ± 1.7
SFT 52k	39.2 ± 1.7	40.7 ± 1.7
SFT 10k	36.7 ± 1.7	44.3 ± 1.7
Binary FeedME	36.6 ± 1.7	37.9 ± 1.7
Quark	35.6 ± 1.7	-
Binary Reward Conditioning	32.4 ± 1.6	-
Davinci001*	24.4 ± 1.5	32.5 ± 1.6
LLaMA 7B*	11.3 ± 1.1	6.5 ± 0.9

Nisan Stiennon & Al, 2020, Learning to summarize from human feedback

Dubois* & Al, 2024, Alpaca Farm: A Simulation Framework for Methods that Learn from Human Feedback

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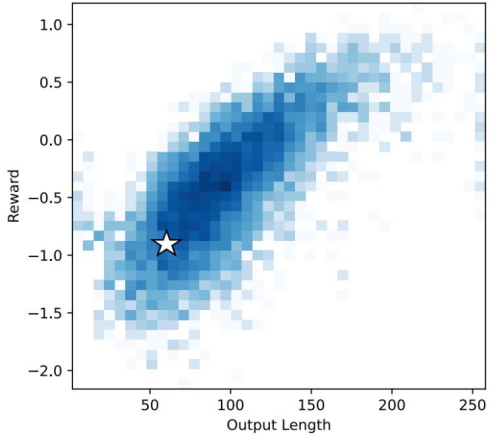
RLHF challenges

- Answer preference is not trivial
- RLHF increases answer size
- Humans do not agree (agree with themselves only 66% of the time)
- Human have lot of variance, model have no variance
  - Ask LLM preferences instead of human preferences



Question: *Why don't adults roll off the bed?*

☆ **SFT (Before); 59 tokens**

Adults typically do not roll off of the bed because they have developed the muscle memory to keep their bodies from involuntarily moving during sleep and maintaining proper posture.

RLHF (After); 243 tokens: **Similar output, but much longer / more details**

Adults generally do not roll off of the bed because they have developed muscle control and awareness which helps them to remain upright while sleeping. Additionally, most adults find it uncomfortable or even painful to move around in their sleep, so rolling off the bed would be difficult without waking up first. There may also be safety concerns such as falling out of bed and potentially hurting oneself if one were to lose balance or fall from the …

▸ **Annotation Guidelines**

Instruction

Tell me about self driving cars

example

Input

AI Response 1

Self driving cars are vehicles that are capable of detecting their surroundings and navigating without any input from a human driver. They use a variety of sensors, such as cameras, radar, and lidar, to detect objects in their environment and use advanced algorithms to make decisions about how to maneuver around them. Self driving cars are becoming increasingly popular and are being used in a variety of applications, such as ride-hailing services and autonomous delivery services.

AI Response 2

Self driving cars are cars that are equipped with sensors, cameras, and software, which allows them to navigate without the need for a driver. Self driving cars are able to monitor their environment and navigate without the assistance of a driver. Self driving cars use a combination of technologies such as GPS, radar, lidar, and computer vision to detect their environment and avoid obstacles. They are also able to update their maps and adjust their routes to account for variables such as traffic and construction.

The main benefit of self driving cars is their ability to reduce crashes and save lives. Autonomous vehicles have the potential to save up to 30,000 lives annually. They are also more energy efficient than traditional vehicles, as they do not require a driver who has to maintain a minimum speed. Autonomous vehicles also create new opportunities for increased mobility, allowing those who are unable to drive to get around.

Rating

- ○ Response 1 is better.
- ○ Response 1 is only slightly better. (only pick this if it's truly close)
- ○ Response 2 is only slightly better. (only pick this if it's truly close)
- ○ Response 2 is better.

Singhal & Al, 2024, A Long Way to Go: Investigating Length Correlations in RLHF

Stanford CS229 I Machine Learning I Building Large Language Models (LLMs) [Youtube]

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Evaluation

How to evaluate a model like Chat GPT ?

- Different methods (DPO, PPO, SFT) can be compared
- Models are not calibrated
- A large diversity of evaluation to cover

Model	Size	Arena Elo	MMLU	Context Window	License
o1-preview	72B +	1355	92.3	128K	Proprietary
ChatGPT-4o-latest	72B +	1335	88.7	128K	Proprietary
Llama-3.1-405b-Instruct	72B +	1266	88.6	128K	Llama 3.1 Community
Llama-3.1-70b-Instruct	40B-72B	1248	86	128K	Llama 3.1 Community
Gemma-2-27b-it	24B-40B	1217	75.2	8K	Gemma license
Llama-3.1-8b-Instruct	4B-24B	1171	73	128K	Llama 3.1 Community
Phi-3-Mini-4k-Instruct	1B-4B	1071	70.9	4K	MIT

Chatbot Arena, Open LM

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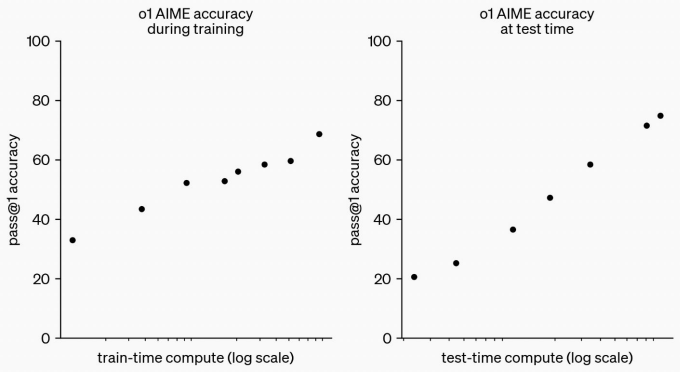
OpenAI o1: “Streaming is dead, long live Chain of Thought”

- Chain of thought (COT)
- Increase test time compute

“Our large-scale reinforcement learning algorithm teaches the model how to think productively using its chain of thought in a highly data-efficient training process. We have found that the performance of o1 consistently improves with more reinforcement learning (train-time compute) and with more time spent thinking (test-time compute). The constraints on scaling this approach differ substantially from those of LLM pretraining, and we are continuing to investigate them.”



o1 performance smoothly improves with both train-time and test-time compute

Open AI, 2024, Learning to Reason with LLMs [Blog]

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OpenAI o1 vs GPT 4o

Prompt:
From the following sentence: Transformers are encoder decoder models
Apply the following steps:
- Create a manim code to display this sentence where each character has a different color
- Iterate through the sentence merging commons pairs as done n the Byte Pair Encoding system
- Change the colors of new pair
- Continue until all commons pair are made
- Update at each step the manim code
- Edit the previous code to not keep one color after merging on the merge pair. The selected color should be the one with the highest number of letters
- Edit the code at the final stage to change color if two adjacent different pair have same color

Open AI o1

GPT 4o

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Conclusion

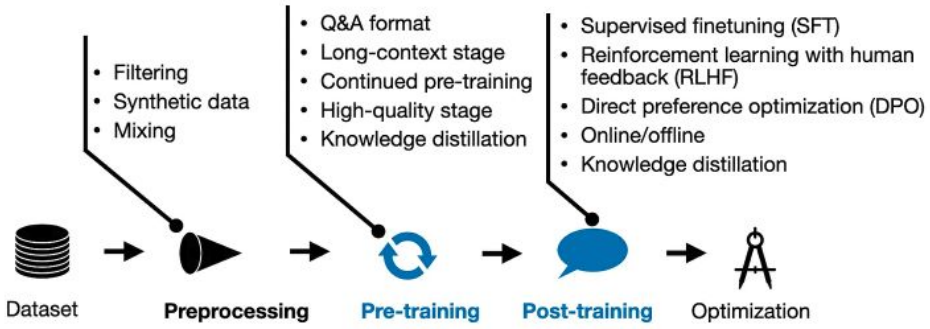
Building GPT 3

- Data preprocessing is a very important step to get quality data
- GPT 3 training consists of two phases (pre and post training)
- Supervised Fine Tuning is the first step of post training phase (ask a human to write the answer)
- RLHF helps the model to align with human preferences
- DPO is the new methods for Alignment, replacing RLHF

General knowledges

Data size and model size depends on compute resources (Scaling Laws, Chinchilla).
OpenAI o1 improves efficiency with longer RLHF training and answer inference time (COT)

"SFT+DPO approach seems to be the most popular preference tuning strategy at the moment due to the ease of use compared to other methods, such as RLHF with PPO."



An overview of the LLM development and training pipeline, with a focus on new pre-training and post-training methodologies discussed in this article

Sebastian Raschka, 2024, New LLM Pre-training and Post-training Paradigms [Blog]

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Conclusion

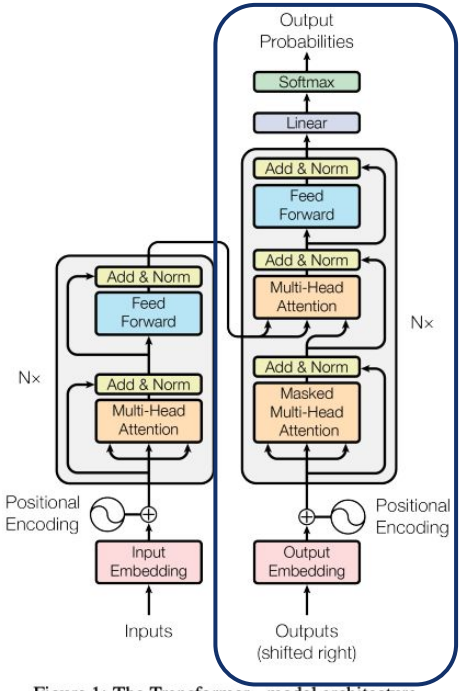
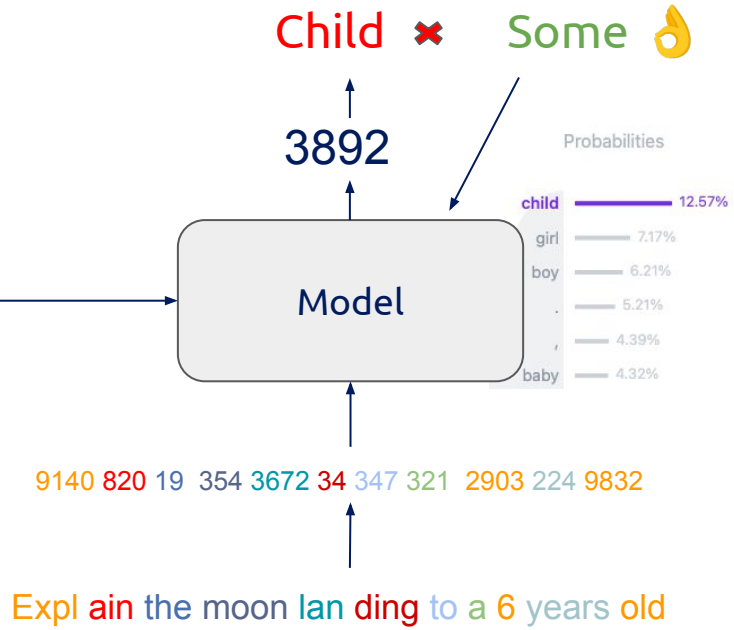


Figure 1: The Transformer - model architecture.



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Quiz Kahoot

Go to : https://kahoot.com/
Click Play
Enter code: XXX XXX
Enter your name

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